Geometric Sequences

Exam code: 9MA0

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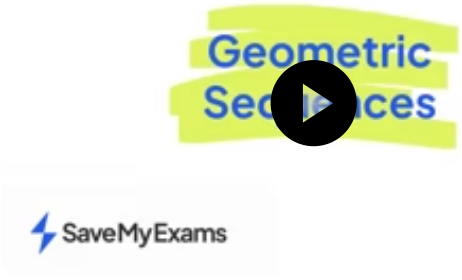

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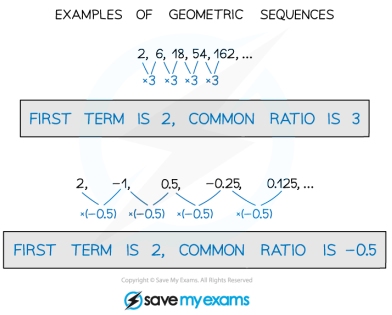
Geometric sequences

What do I need to know about geometric sequences?

- In a **geometric sequence**, there is a **common ratio** between consecutive terms in the sequence



- You need to know the nth **term formula** for a geometric sequence

$$u_n = ar^{n-1}$$

  - a is the first term
  - r is the common ratio

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

If you know two terms in a geometric sequence you can find ***a*** and ***r*** using simultaneous equations.

e.g. THE 6th TERM OF A GEOMETRIC SEQUENCE IS 486.
THE 10th TERM IS 39 366.
FIND THE FIRST TERM AND THE COMMON DIFFERENCE.

$u_n = ar^{n-1}$

$ar^5 = 486$ ①
$ar^9 = 39\,366$ ②

SOLVE THESE SIMULTANEOUS EQUATIONS

$r^4 = \frac{39\,366}{486} = 81$

② DIVIDED BY ①

$r = 81^{\frac{1}{4}} = 3$

$a(3)^5 = 486$
$243a = 486$
$a = 2$

SUBSTITUTE INTO ①

THE FIRST TERM IS 2 AND
THE COMMON RATIO IS 3

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Worked Example

The first three terms in a geometric sequence are $2 - x$, $3x$, and x^2 respectively where $x < 0$.

a) Show that $x^3 + 7x^2 = 0$.

b) Find the 6th term of the sequence.

a) THIS IS A GEOMETRIC SEQUENCE SO
THERE'S A SINGLE COMMON RATIO

$r = \frac{3x}{2-x}$ — 2nd TERM DIVIDED BY 1st TERM

AND

$r = \frac{x^2}{3x}$ — 3rd TERM DIVIDED BY 2nd TERM

SO $\frac{3x}{2-x} = \frac{x^2}{3x}$

$(3x)^2 = x^2(2-x)$
$9x^2 = 2x^2 - x^3$
$x^3 + 7x^2 = 0$

b) $x^2(x+7) = 0$ — SOLVE THE EQUATION FROM a)

$x = 0$ OR $x = -7$
BUT $x < 0$ SO $x = -7$

$a = 2 - (-7) = 9$

$r = \frac{3(-7)}{2-(-7)} = \frac{-21}{9} = \frac{-7}{3}$

FIND a AND r

$u_6 = 9\left(\frac{-7}{3}\right)^5 = -\frac{16\,807}{27}$

$u_n = ar^{n-1}$

THE SIXTH TERM IS $-\frac{16\,807}{27}$

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